01 — Yield Curve Exploration

ARCM: Adaptive Rate Curve Model | Liyuan Zhang

Explores the Nelson-Siegel yield curve decomposition produced by `pipeline.py`.

Prerequisite: `python pipeline.py` from the `arcm/` directory.

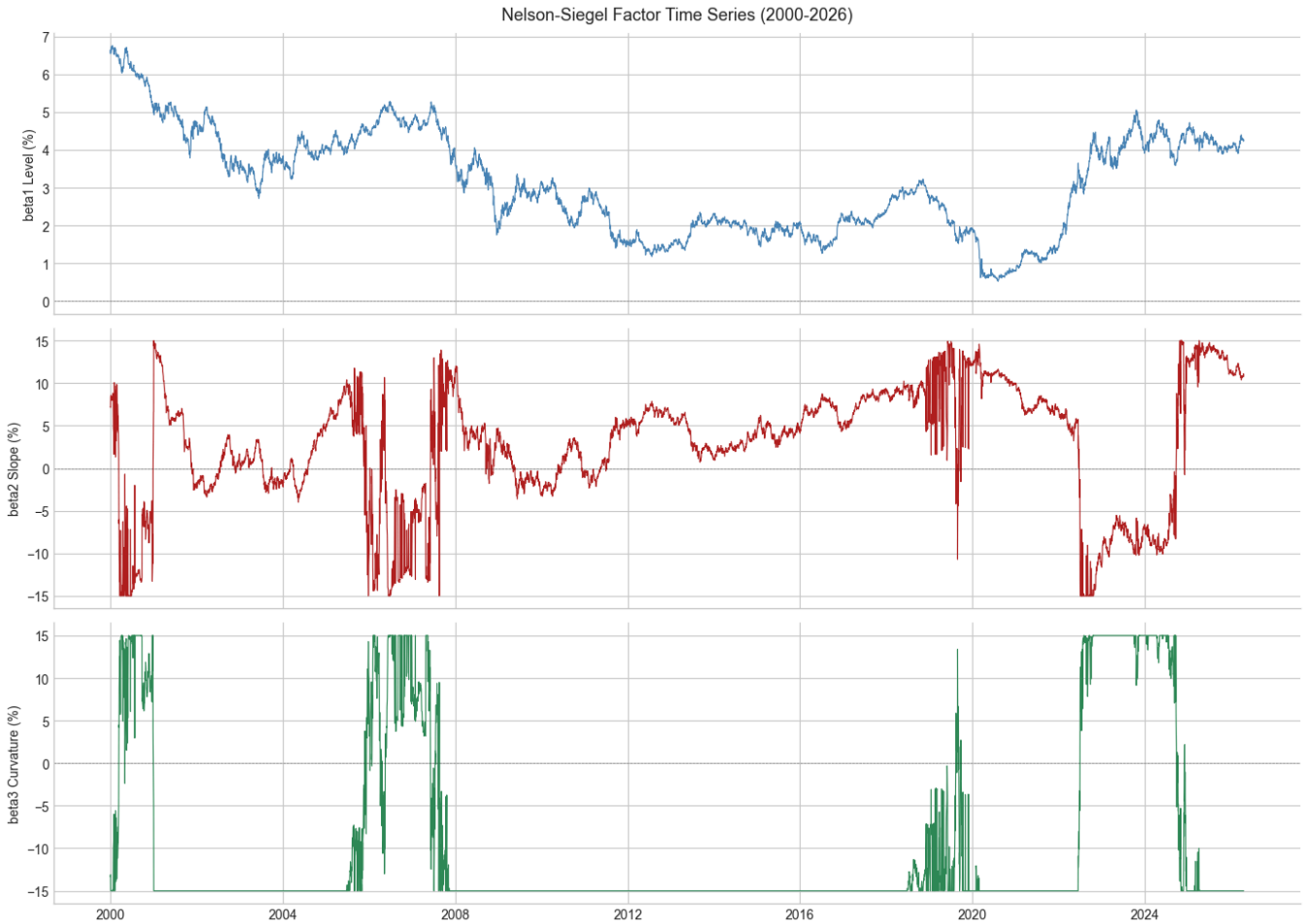
1. Load pipeline outputs

```
Factors: 6575 rows  (2000-01-03 to 2026-04-16)
```

	date	beta1	beta2	beta3	y1y	slope_2s10s	curve_shape
6570	2026-04-10	4.279032	10.739537	-15.0	4.019571	0.103785	Flat
6571	2026-04-13	4.263185	10.914710	-15.0	4.014392	0.099518	Flat
6572	2026-04-14	4.229047	11.095647	-15.0	3.991273	0.095110	Flat
6573	2026-04-15	4.250261	10.952798	-15.0	4.003788	0.098590	Flat
6574	2026-04-16	4.272311	10.773613	-15.0	4.014925	0.102955	Flat

2. Nelson-Siegel factors over time

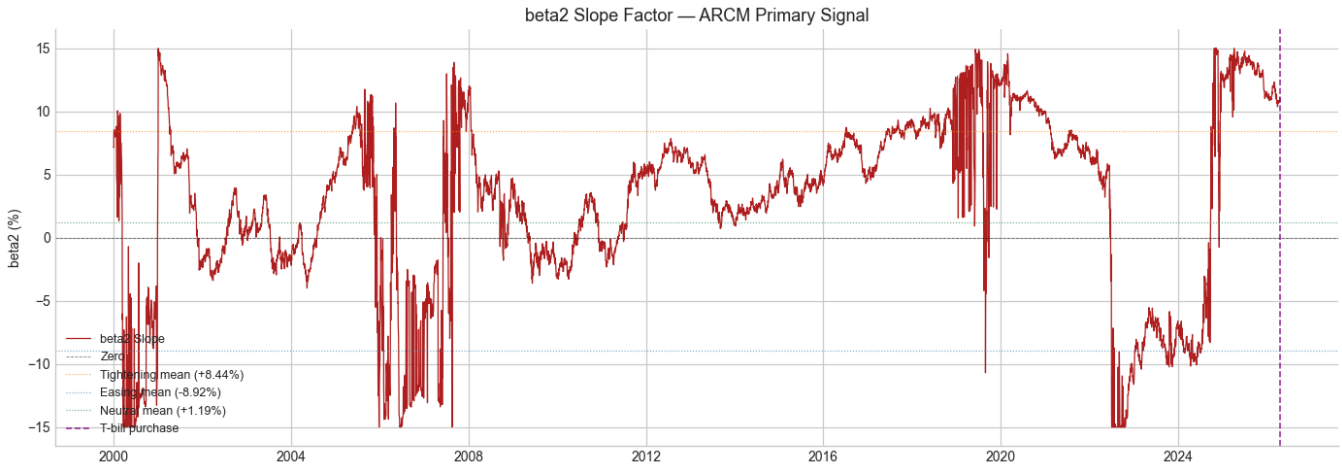
- **beta1 (Level):** long-run rate anchor
- **beta2 (Slope):** short-end premium — negative means inverted curve
- **beta3 (Curvature):** medium-term hump — captures policy uncertainty



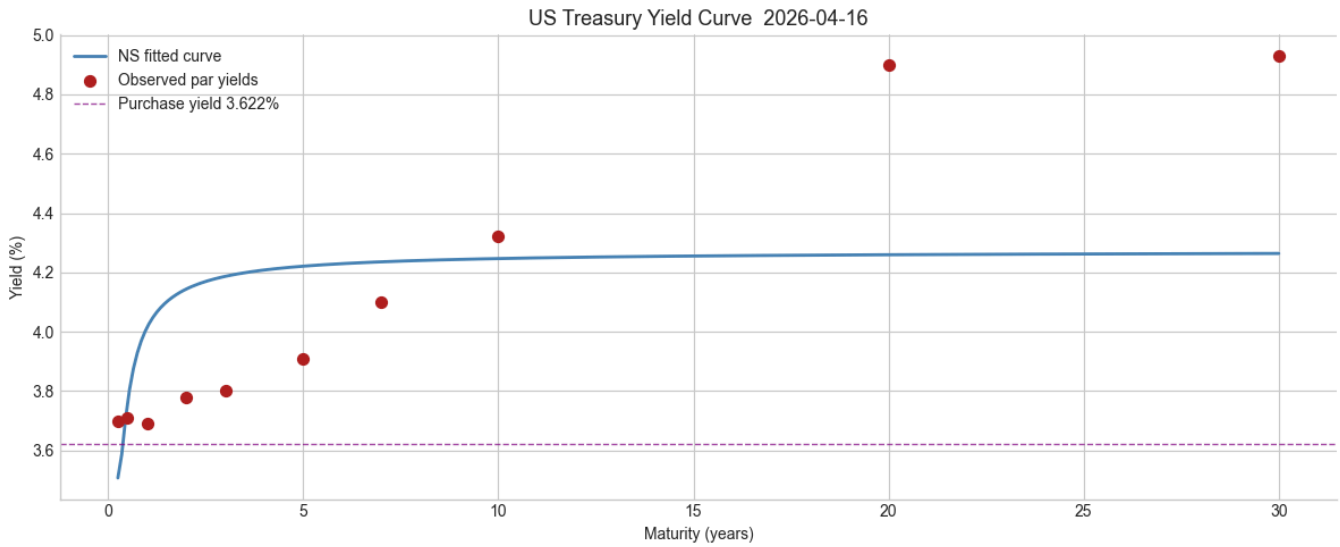
3. Slope factor (beta2) — the ARCM primary signal

beta2 is the core input to regime detection and OU calibration.

- Prolonged negative beta2 during ZLB era (2009-2015): mean-reversion breaks down
- Sharp negative excursion during 2022-2023 hiking cycle
- Current reading (+10.77%) well above tightening regime mean (+8.44%)



4. Current yield curve snapshot

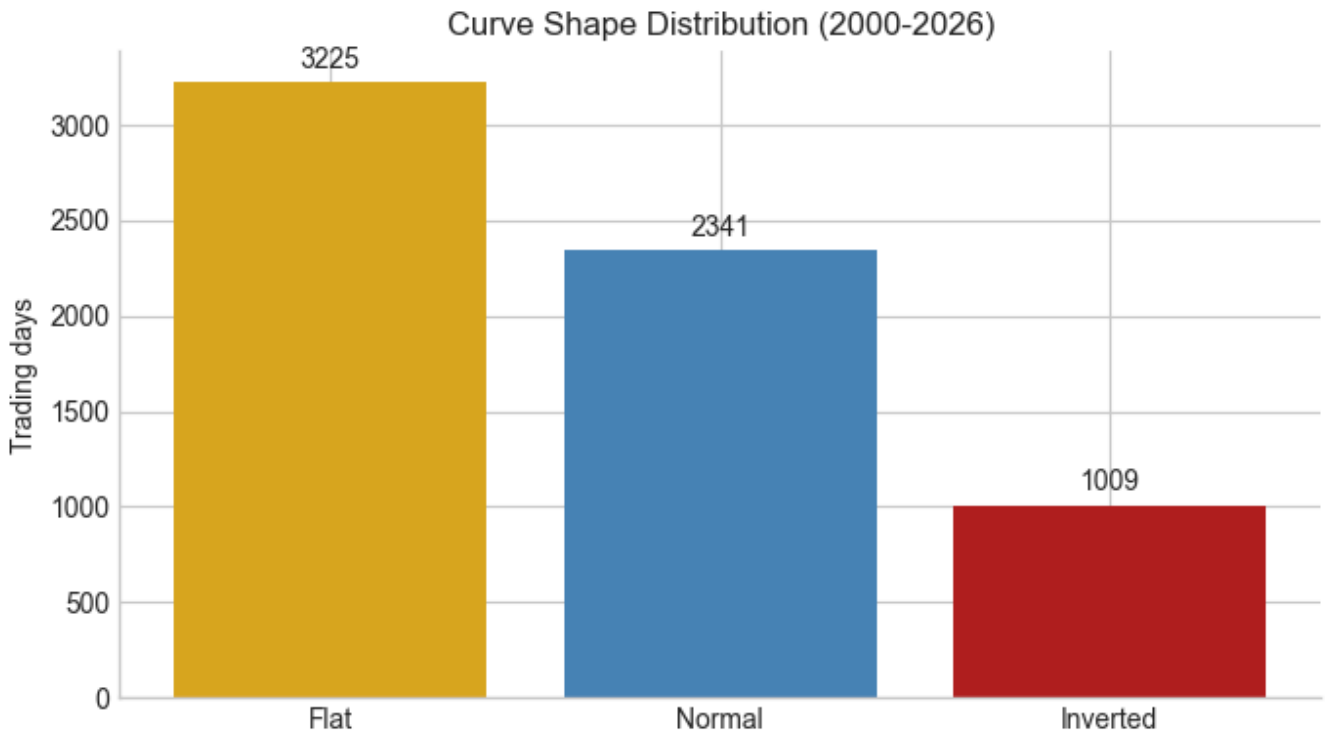



```
beta1=4.272%  beta2=10.774%  beta3=-15.000%
1yr NS yield: 4.015%  |  2s10s: 0.103%  |  Shape: Flat
```

5. NS fit quality

```
Median RMSE     : 50.2 bps
95th pct RMSE   : 113.2 bps
Poor fit days   : 4971  (>20 bps)
```

6. Curve shape distribution




```
Flat        3225 days  (49.0%)
Normal      2341 days  (35.6%)
Inverted    1009 days  (15.3%)
```